TASK 23- JENKINS INSTALLATION AND AUTOMATION SERVER SETUP ON AMAZON LINUX

Objective

1. Provision an Amazon Linux EC2 instance and configure network security rules for remote web access.
2. Install and configure Java 21 (Amazon Corretto) as the required runtime environment for modern Jenkins builds.
3. Add the official stable Jenkins RPM repository and import GPG verification keys.
4. Install the Jenkins package and manage its lifecycle using systemd.
5. Retrieve the initial administrator password and unlock the web management dashboard.

1. Environment Preparation & System Updates

The local package manager index was updated on the Amazon Linux EC2 instance to ensure system libraries and dependencies were current.

Steps:

1. Connect to the EC2 instance via SSH or EC2 Instance Connect.
2. Execute the system update command:

Bash

```
sudo dnf update -y
```

2. Java 21 Installation and Configuration

Because modern Jenkins builds require Java 21, the enterprise-grade Amazon Corretto JDK package was installed.

Steps:

1. Install Java 21 and fontconfig dependencies:

Bash

```
sudo dnf install -y fontconfig java-21-amazon-corretto
```

```
[ec2-user@ip-172-31-12-139 ~]$ sudo dnf install -y fontconfig java-21-amazon-corretto
Last metadata expiration check: 0:00:10 ago on Wed Sep 30 07:18:40 2026.
Dependencies resolved.
================================================================================
 Package                    Arch      Version                    Repository   Size
================================================================================
Installing:
 fontconfig                 x86_64    2.13.94-2.amzn2023.0.2     amazonlinux  273 k
 java-21-amazon-corretto    x86_64    1:21.0.12+9-1.amzn2023.1   amazonlinux  217 k
Installing dependencies:
 alsa-lib                   x86_64    1.2.7.2-1.amzn2023.0.4     amazonlinux  503 k
 cairo                      x86_64    1.18.0-4.amzn2023.0.3      amazonlinux  717 k
 dejavu-sans-fonts          noarch    2.37-16.amzn2023.0.2       amazonlinux  1.3 M
 dejavu-sans-mono-fonts     noarch    2.37-16.amzn2023.0.2       amazonlinux  467 k
 dejavu-serif-fonts         noarch    2.37-16.amzn2023.0.2       amazonlinux  1.0 M
 fonts-filesystem           noarch    1:2.0.5-12.amzn2023.0.2    amazonlinux  9.5 k
```

2. Configure the system Java default using the alternatives manager:

Bash
sudo alternatives --config java

(Selected the number corresponding to Java 21 and pressed Enter).

```
Complete!
[ec2-user@ip-172-31-12-139 ~]$ sudo alternatives --config java

There is 1 program that provides 'java'.

  Selection    Command
-----------------------------------------------
*+ 1           /usr/lib/jvm/java-21-amazon-corretto.x86_64/bin/java

Enter to keep the current selection[+], or type selection number: 1
```

3. Jenkins Repository and GPG Key Configuration

The official stable Jenkins repository configuration file and GPG verification key were added to the system package manager.

Steps:

1. Download and save the Jenkins repository configuration file:

Bash

sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.re

```
[ec2-user@ip-172-31-12-139 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2026-09-30 07:19:32--  https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
--2026-09-30 07:19:33--  https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins 100%[=================================>]     267  --.-KB/s    in 0s

2026-09-30 07:19:33 (15.3 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
```

2. Import the official security signing key to verify package integrity:

Bash
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key

4. Jenkins Installation and Service Management

The Jenkins automation server package was installed and configured to run automatically as a background daemon.

Steps:

1. Install the Jenkins package via DNF:

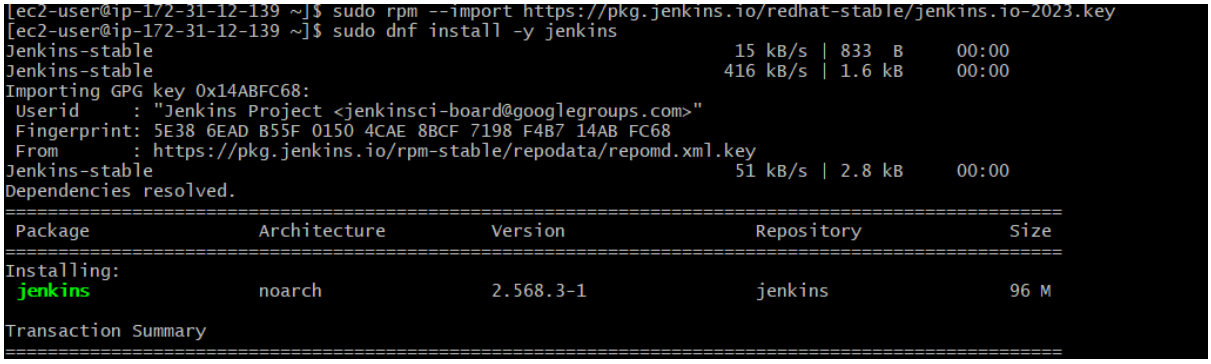
Bash

sudo dnf install -y jenkins

2. Refresh the systemd manager configuration cache, start Jenkins, and enable automatic startup on boot:

Bash

sudo systemctl daemon-reload
sudo systemctl enable --now Jenkins



5. Security Group Configuration and Dashboard Unlock

Network access was verified via the AWS security group, and the initial setup token was retrieved to unlock the web interface.

Steps:

1. Ensure the EC2 Security Group has an inbound rule allowing **Custom TCP on Port 8080** from your IP address or anywhere (0.0.0.0/0).
2. Navigate to the Jenkins web interface in a browser:

Plaintext

http://<your-ec2-public-ip>:8080

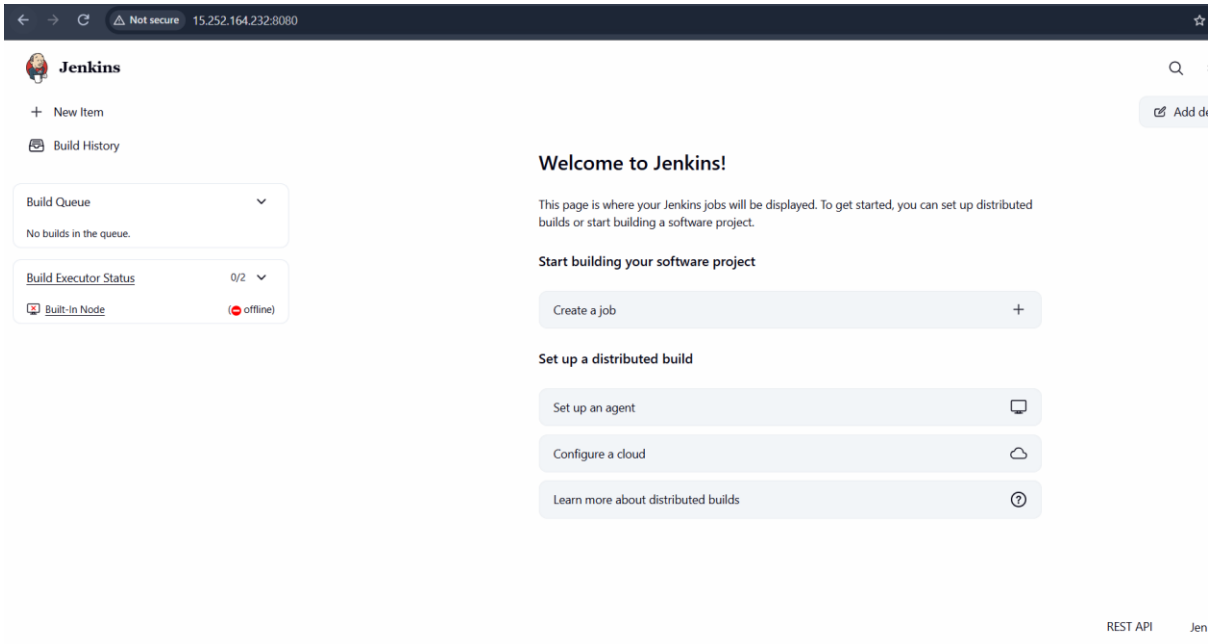
3. Retrieve the initial administrator password from the server terminal:

Bash

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

4. Paste the password into the browser prompt, select **Install suggested plugins**, and create the administrator user account to complete the setup.

```
[ec2-user@ip-172-31-12-139 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
2f1989c52e65407b90cae7b036643a6e
[ec2-user@ip-172-31-12-139 ~]$
```



Conclusion

This task successfully demonstrated the complete lifecycle of setting up an automation server: updating system packages, installing and configuring Java 21 runtime environments, adding official package repositories securely, managing background services via systemd, and unlocking the Jenkins web management console.